

CHEM 3PC3 - Mathematical Tools for Chemical Problems

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Comments about the quiz

- ▶ Having a function $f(t) = \sqrt{t^3}$ the derivative w.r.t t is:

$$\frac{df(t)}{dt} \neq \frac{dt}{dx} \neq \frac{df(t)}{dx} \quad (1)$$

- ▶ Are these the same?

$$\frac{df}{dt} \quad \text{vs.} \quad \frac{\partial f}{\partial t} \quad \text{vs.} \quad \frac{\delta F}{\delta f(t)}$$

- ▶ $\frac{df}{dt}$: **total derivative**, counts every way f depends on t .
- ▶ $\frac{\partial f}{\partial t}$: **partial derivative**, all other variables held fixed. For $f(x(t), t)$,

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} \frac{dx}{dt}.$$

If f depends only on t , then $\frac{\partial f}{\partial t} = \frac{df}{dt}$.

- ▶ $\frac{\delta F}{\delta f(t)}$: **functional derivative**, the derivative of a functional $F[f]$ with respect to a whole function
- ▶ In thermodynamics, δq and δw are **inexact differentials**: path-dependent, not differentials of a state function.

Limits

- ▶ The function $1/x$ is not defined everywhere. It blows up at $x = 0$ where we divide by zero.
- ▶ Sometimes we can save a function from being undefined at some point.
- ▶ A silly example is $f(x) = x^2/x$. Which is initially not defined at $x = 0$.
- ▶ But we can see that $f(x) = x$.
- ▶ This is done by dividing by a common factor. We can see some examples.

Limits

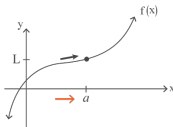
- ▶ The function $f(x) = (x^3 - 1)/(x - 1)$ is not defined at $x = 1$.
- ▶ But for x really close to 1 nothing bad happens.
- ▶ We can evaluate the function at points closer and closer to 1 and get closer and closer to a value.
- ▶ Using the difference of two cubes
 $(x^3 - 1^3) = (x - 1)(x^2 + x + 1)$.

$$f(x) = \frac{(x - 1)(x^2 + x + 1)}{(x - 1)} = x^2 + x + 1 \quad (2)$$

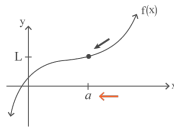
- ▶ And now when $x = 1$, there is a natural value associated with the function.
- ▶ Definition: We write $x \rightarrow a$ to indicate that x approaches a .

Limits

- ▶ Why do we worry about limits ? We will define derivatives and integrals using limits.
- ▶ The limit can be from either side, from the left $x \rightarrow a^-$ or from the right $x \rightarrow a^+$.



$$\lim_{x \rightarrow a^-} f(x) = \text{left-sided limit}$$



$$\lim_{x \rightarrow a^+} f(x) = \text{right-sided limit}$$

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- ▶ Example, $f(x) = x^2$
- ▶ A function $f(x)$ has a limit at a point a if exists a unique b such that:

$$\lim_{x \rightarrow a} f(x) = b$$

Rate of Change

- ▶ Given a function f and a constant $h > 0$, we can look at a new function:

$$Df(x) = \frac{f(x+h) - f(x)}{h}. \quad (4)$$

- ▶ Is the average rate of change of the function with step size h .
- ▶ When we take the limit $h \rightarrow 0$ we have the instantaneous rate of change.
- ▶ And we can derive important formulas: $\frac{d}{dx}x^n = nx^{n-1}$,
 $\frac{d}{dx}\exp(ax) = a\exp(x)$, $\frac{d}{dx}\sin(ax) = a\cos(ax)$,
 $\frac{d}{dx}\cos(ax) = -a\sin(ax)$.
- ▶ Definition: If the limit $\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exist, we say f is differentiable at point x .

Continuity

- ▶ A function f is continuous at a point x_0 if a value of $f(x_0)$ can be found such that $f(x) \rightarrow f(x_0)$ for $x \rightarrow x_0$.
- ▶ Intuitively, a function is continuous if one can draw the graph of the function without lifting the pencil.
- ▶ Why do we like continuity ?
- ▶ Continuity also will be useful when finding maxima and minima.
- ▶ Finding discontinuities and plotting it is a work we can do in Python.

Graphs

- ▶ In Python, common practice is to use matplotlib for graphics.
- ▶ In matplotlib, a plot consists of a figure and one or more axes.
- ▶ Example: Determine for the function $f(x) = \log(|x^2 - 1|)$ where the discontinuities appear.
- ▶ This can be found by knowing that the log is only defined for positive values $x > 0$.
- ▶ So the discontinuities will show up when we evaluate log at zero i.e. $|x^2 - 1| = 0$.
- ▶ Which leads to two solutions $x = 1$ and $x = -1$.
- ▶ We can actually **plot** the function to see if our analysis hold.

Exercises

- ▶ Determine for each of the following functions, where discontinuities appear:
 - ▶ $f(x) = \sin(\cos(\pi/x))$
 - ▶ $f(x) = \cot(x) + \tan(x) + x^4$
 - ▶ $f(x) = (x^2 + 2x + 1)/(x + 10 + (x - 1)^2/(x - 1))$
 - ▶ $f(x) = (x^2 - 4x)/x$
- ▶ Evaluate:
 - ▶ $\lim_{z \rightarrow 8} \frac{2z^2 - 17z + 8}{8 - z}$
 - ▶ $\lim_{x \rightarrow 0} \frac{x}{3 - \sqrt{x+9}}$

1. Given the function

$$f(x) = \begin{cases} 7 - 4x, & x < 1, \\ x^2 + 2, & x \geq 1. \end{cases}$$

Evaluate the following limits, if they exist.

- (a) $\lim_{x \rightarrow -6} f(x)$
- (b) $\lim_{x \rightarrow 1} f(x)$

Numerical integration

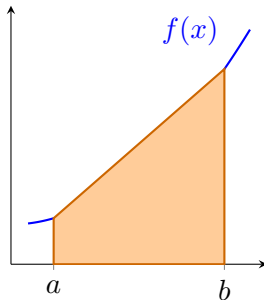
- ▶ Many integrals in chemistry have no closed-form antiderivative, or we only know the integrand as data on a grid (e.g. $C_P(T)$ measured at a few temperatures).
- ▶ The idea of numerical integration is to approximate the area under $f(x)$ with simple shapes whose area we know.
- ▶ The simplest useful shape is a **trapezoid**: connect neighbouring points with straight lines.
- ▶ This is the **trapezoidal rule**, and it is one line of code in Python.

The trapezoidal rule: one interval

- ▶ Replace $f(x)$ on $[a, b]$ by the straight line joining $(a, f(a))$ and $(b, f(b))$.
- ▶ The area under that line is a trapezoid,

$$\int_a^b f(x) dx \approx \frac{b-a}{2} (f(a) + f(b)).$$

- ▶ It is the width times the average height.

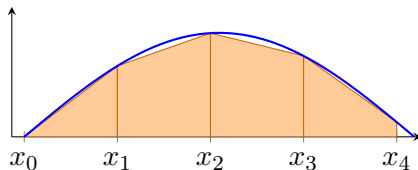


The composite trapezoidal rule

- ▶ One trapezoid is too crude, so we split $[a, b]$ into $N - 1$ intervals of width $h = \frac{b-a}{N-1}$, with grid points $x_i = a + ih$.
- ▶ Adding the area of every trapezoid gives

$$\begin{aligned}\int_a^b f(x) dx &\approx \frac{h}{2} \sum_{i=0}^{N-2} \left(f(x_i) + f(x_{i+1}) \right) \\ &= h \left[\frac{1}{2}f_0 + f_1 + \cdots + f_{N-2} + \frac{1}{2}f_{N-1} \right].\end{aligned}$$

- ▶ Interior points are shared by two trapezoids, so they get weight 1. The two end points get weight $\frac{1}{2}$.



How accurate is it?

- ▶ The error of the composite rule is

$$E = -\frac{(b-a)h^2}{12} f''(\xi), \quad \text{for some } \xi \in [a, b].$$

- ▶ The error scales as h^2 : halving the grid spacing reduces the error by a factor of 4. We say the method is **second order**.
- ▶ The rule is **exact** for straight lines, since $f'' = 0$.
- ▶ If $f'' > 0$ (the curve bends upwards) the trapezoids sit above the curve and *overestimate* the integral. If $f'' < 0$ they *underestimate* it.
- ▶ Steep regions (large f'') need a finer grid, so check convergence by repeating the calculation with more points.
- ▶ Implement $\int_a^b f(x)dx \approx \frac{h}{2} \sum_{i=0}^{N-2} (f(x_i) + f(x_{i+1}))$ in python.

The trapezoidal rule in Python

- ▶ With NumPy arrays, the whole sum is one line:

```
import numpy as np

def trapezoidal_rule(f, h):
    # f: integrand on a uniform grid, h: spacing
    return h / 2 * np.sum(f[:-1] + f[1:])

x = np.linspace(0, np.pi, 11)    # 11 points
h = x[1] - x[0]
print(trapezoidal_rule(np.sin(x), h))    # 1.98352
print(np.trapezoid(np.sin(x), x))        # NumPy
```

- ▶ The exact answer is $\int_0^\pi \sin x \, dx = 2$. With only 11 points the error is already below 1%.
- ▶ The [notebook](#) shows the trapezoids, a convergence test, and a chemistry example.